

ENERGY: sMAPE at each forecast step				
Lookback = 168 steps, Forecast horizon = 72 steps				
Forecast step	CNN	LSTM	dCeNN+ELM	dCeNN+ELM+ASP
1.000	57.603	71.949	77.262	55.502
2.000	59.242	67.368	77.374	55.561
3.000	60.783	65.529	77.861	56.046
4.000	62.019	65.159	78.032	56.238
5.000	63.406	65.407	78.118	56.333
6.000	64.156	65.854	78.576	56.812
7.000	65.088	66.354	78.827	57.055
8.000	65.642	66.894	78.804	56.996
9.000	66.295	67.380	78.715	56.850
10.000	66.766	67.810	78.580	56.770
11.000	67.042	68.132	78.299	56.536
12.000	67.645	68.367	78.611	56.850
13.000	67.544	68.591	78.844	57.082
14.000	67.649	68.890	79.040	57.281
15.000	68.281	69.029	79.640	57.879
16.000	68.859	69.176	80.344	58.583
17.000	69.014	69.331	80.336	58.574
18.000	68.880	69.514	79.945	58.180
19.000	68.972	69.649	79.543	57.777
20.000	69.165	69.742	78.971	57.197
21.000	69.089	69.759	78.593	56.822
22.000	69.239	69.661	78.492	56.712
23.000	69.296	69.615	78.487	56.701
24.000	68.739	69.785	78.210	56.445
25.000	69.038	70.023	77.792	55.923
26.000	69.683	70.220	77.549	55.642
27.000	69.731	70.339	77.623	55.774
28.000	69.841	70.400	77.722	55.958
29.000	69.989	70.423	77.548	55.781
30.000	69.759	70.312	77.270	55.515
31.000	69.982	70.307	77.221	55.467
32.000	70.034	70.393	77.400	55.634
33.000	70.159	70.502	77.797	56.015
34.000	70.078	70.585	78.177	56.419
35.000	70.073	70.564	78.175	56.415
36.000	69.990	70.511	78.390	56.623
37.000	69.915	70.449	78.665	56.897
38.000	69.958	70.436	78.887	57.132
39.000	69.955	70.416	79.474	57.719
40.000	70.200	70.404	80.162	58.404
41.000	70.296	70.385	80.384	58.624
42.000	69.949	70.376	80.012	58.250
43.000	70.103	70.402	79.626	57.862
44.000	70.304	70.384	79.208	57.443
45.000	70.409	70.318	78.864	57.099
46.000	70.313	70.226	78.535	56.766
47.000	70.187	70.208	78.209	56.457
48.000	70.041	70.306	77.740	56.004
49.000	70.472	70.482	77.206	55.474
50.000	70.489	70.621	77.139	55.394
51.000	70.464	70.717	77.393	55.642
52.000	70.467	70.723	77.903	56.139
53.000	70.383	70.742	78.199	56.429
54.000	70.422	70.727	78.218	56.456
55.000	70.687	70.739	77.907	56.141
56.000	70.523	70.785	77.392	55.602
57.000	70.546	70.870	77.100	55.279
58.000	70.757	70.942	77.181	55.358
59.000	70.166	70.973	77.343	55.565
60.000	70.296	71.008	77.594	55.793
61.000	70.356	71.031	77.936	56.165
62.000	70.381	71.056	78.207	56.440
63.000	70.181	71.037	78.634	56.875
64.000	70.594	71.010	79.168	57.407
65.000	70.431	71.039	79.157	57.395
66.000	70.555	71.081	78.834	57.070
67.000	70.329	71.102	78.529	56.763
68.000	70.295	71.105	78.408	56.641
69.000	70.711	71.082	78.569	56.802
70.000	70.644	71.039	78.803	57.041
71.000	70.663	71.061	78.760	57.006
72.000	70.331	71.276	78.390	56.647

• This table shows how prediction error changes from the first forecast step to the last forecast step.
• It is especially useful for explaining whether a model degrades smoothly or sharply as it predicts further into the future.
• Lower is better. This is a percentage-style error that is easier to compare across scales.